

Tab 1

```

#include <stdio.h>
#include <stdlib.h>

#define MAX 5

int queue[MAX];
int front = -1, rear = -1;

/* Function declarations */
void enqueue();
void dequeue();
void peek();
void display();

int main()
{
    int choice;

    while(1)
    {
        printf("\n--- Queue Menu ---\n");
        printf("1. Enqueue(insert)\n");
        printf("2. Dequeue(delete)\n");
        printf("3. Peek\n");
        printf("4. Display\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1: enqueue(); break;
            case 2: dequeue(); break;
            case 3: peek(); break;
            case 4: display(); break;
            case 5: exit(0);
            default: printf("Invalid choice\n");
        }
    }
}

/* Enqueue operation */
void enqueue()
{

```

```

    int item;

    if(rear == MAX - 1)
    {
        printf("Queue Overflow\n");
        return;
    }

    printf("Enter element to insert: ");
    scanf("%d", &item);

    if(front == -1)
        front = 0;

    rear++;
    queue[rear] = item;

    printf("Element inserted successfully\n");
}

/* Dequeue operation */
void dequeue()
{
    if(front == -1 || front > rear)
    {
        printf("Queue Underflow\n");
        return;
    }

    printf("Deleted element: %d\n", queue[front]);
    front++;

    if(front > rear)
    {
        front = rear = -1;
    }
}

/* Peek operation */
void peek()
{
    if(front == -1)
    {
        printf("Queue is empty\n");
    }
}

```

```
        return;
    }

    printf("Front element: %d\n", queue[front]);
}

/* Display operation */
void display()
{
    int i;

    if(front == -1)
    {
        printf("Queue is empty\n");
        return;
    }

    printf("Queue elements:\n");
    for(i = front; i <= rear; i++)
    {
        printf("%d ", queue[i]);
    }
    printf("\n");
}
```

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Terminal

```
que --- Queue Menu ---
88 1. Enqueue(insert)
89 2. Dequeue(delete)
90 3. Peek
91 4. Display
92 5. Exit
93 Enter your choice: 1
94 Enter element to insert: 2
95 Element inserted successfully
96
97 --- Queue Menu ---
98 1. Enqueue(insert)
99 2. Dequeue(delete)
100 3. Peek
101 4. Display
102 5. Exit
103 Enter your choice: 2
104 Deleted element: 2
105
106 --- Queue Menu ---
107 1. Enqueue(insert)
108 2. Dequeue(delete)
109 3. Peek
110
111
112
113
Compiler Compilation finished successfully.
line: 113 / 113 col: 0 sel: 0 INS TAB EOL: LF encoding: UTF-8 filetype: C scope: unknown
```

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Terminal

```
1. Enqueue(insert)
2. Dequeue(delete)
88 3. Peek
89 4. Display
90 5. Exit
91 Enter your choice: 2
92 Deleted element: 2
93
94 --- Queue Menu ---
95 1. Enqueue(insert)
96 2. Dequeue(delete)
97 3. Peek
98 4. Display
99 5. Exit
100 Enter your choice: 3
101 Queue is empty
102
103 --- Queue Menu ---
104 1. Enqueue(insert)
105 2. Dequeue(delete)
106 3. Peek
107 4. Display
108 5. Exit
109 Enter your choice: 
110
111
112
113
Compiler Compilation finished successfully.
line: 113 / 113 col: 0 sel: 0 INS TAB EOL: LF encoding: UTF-8 filetype: C scope: unknown
```

Tab 2

